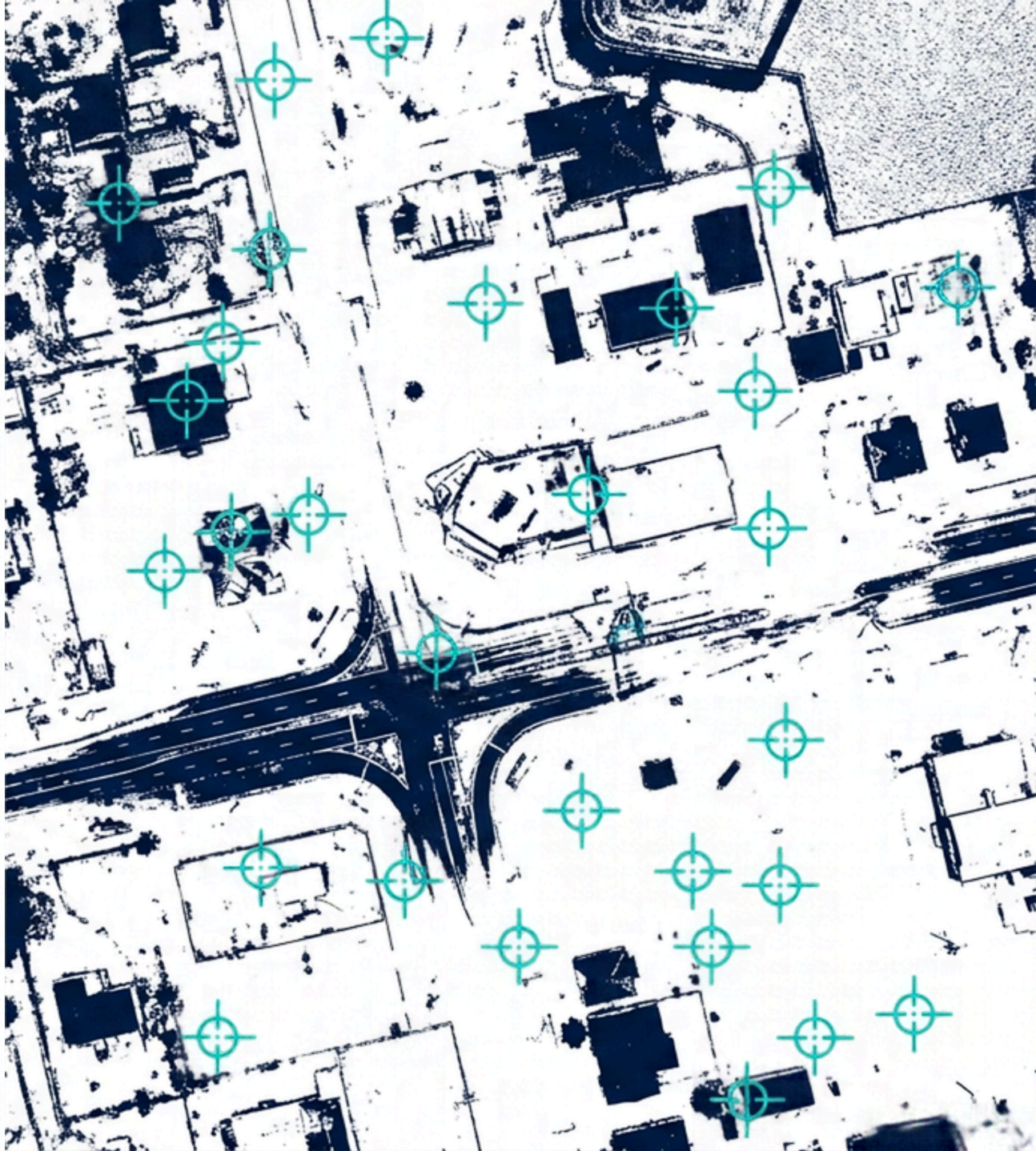
SkyRescue

AI-Enabled Disaster Response Decision Support & Management Information System

System Architecture & Engineering Blueprint



SKYRESCUE DSS: SYSTEM WORKFLOW COMPARISON

	Traditional Response	SkyRescue DSS
Data Intake & Triage	Manual review of surging field images; high risk of missed incidents.	Automated Layer 1 validation (EXIF, GPS, 30m freshness, SHA-256 deduplication).
Hazard Prioritization	Subjective, slow queueing based on human estimation.	AI Severity Scoring [0-100] + AHP/TOPSIS mathematical ranking.
Resource Allocation	Manual map cross-referencing and availability checks.	Multi-criteria Team Match Scoring (Specialty, distance, capacity).
Command Decision	Fragmented tools.	SAR Coordinator verified dispatch with immutable SQLite WAL audit trails.

SKYRESCUE DSS: OPERATIONAL RULES & LIFECYCLE

Human-in-the-Loop

AI Proposes, Coordinator Disposes.

100% of incident dispatches require explicit SAR Coordinator verification.

Zero automated closed-loop dispatches.

Complete Review

Every valid image enters the queue.

Normal cases with P > 0.90 bypass casualty scoring directly to human sign-off.

Architectural Scope

2-Portal Architecture.

Fast, local execution

No physical UAV control; no live team GPS tracking.

Incident Lifecycle



Context Diagram (v3)

Actors

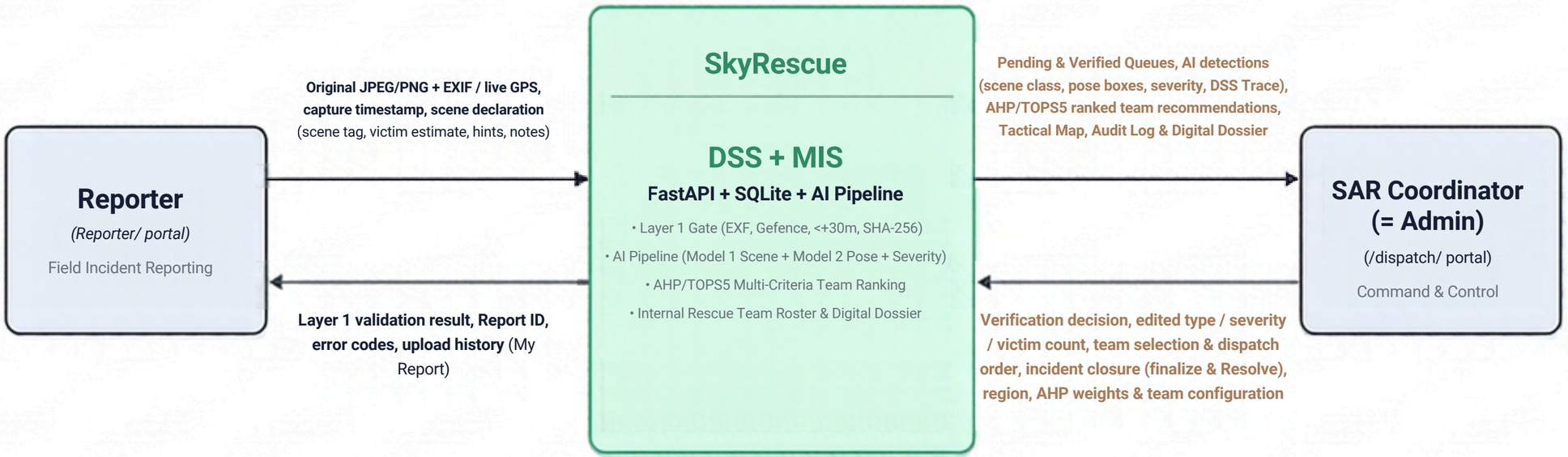
Simplified 2-Actor Model. Field Reporters submit constraints (35MB max, EXIF/GPS), SAR Coordinator commands the digital dossier.

Core Stack

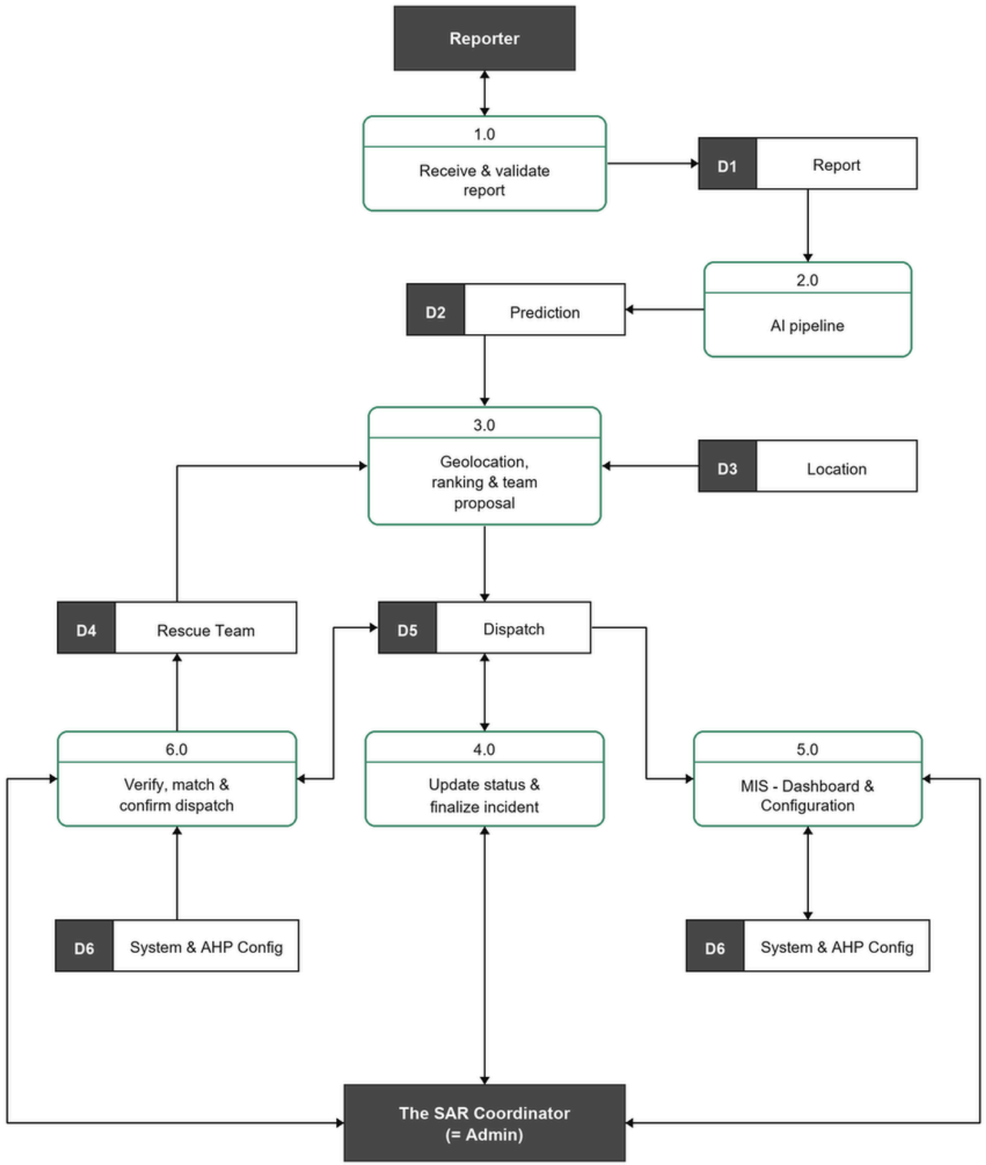
FastAPI Backend + SQLite (WAL mode) persistence layer.

Sync Bus

Real-time Server-Sent Events (SSE) with 2-second automated polling failover.



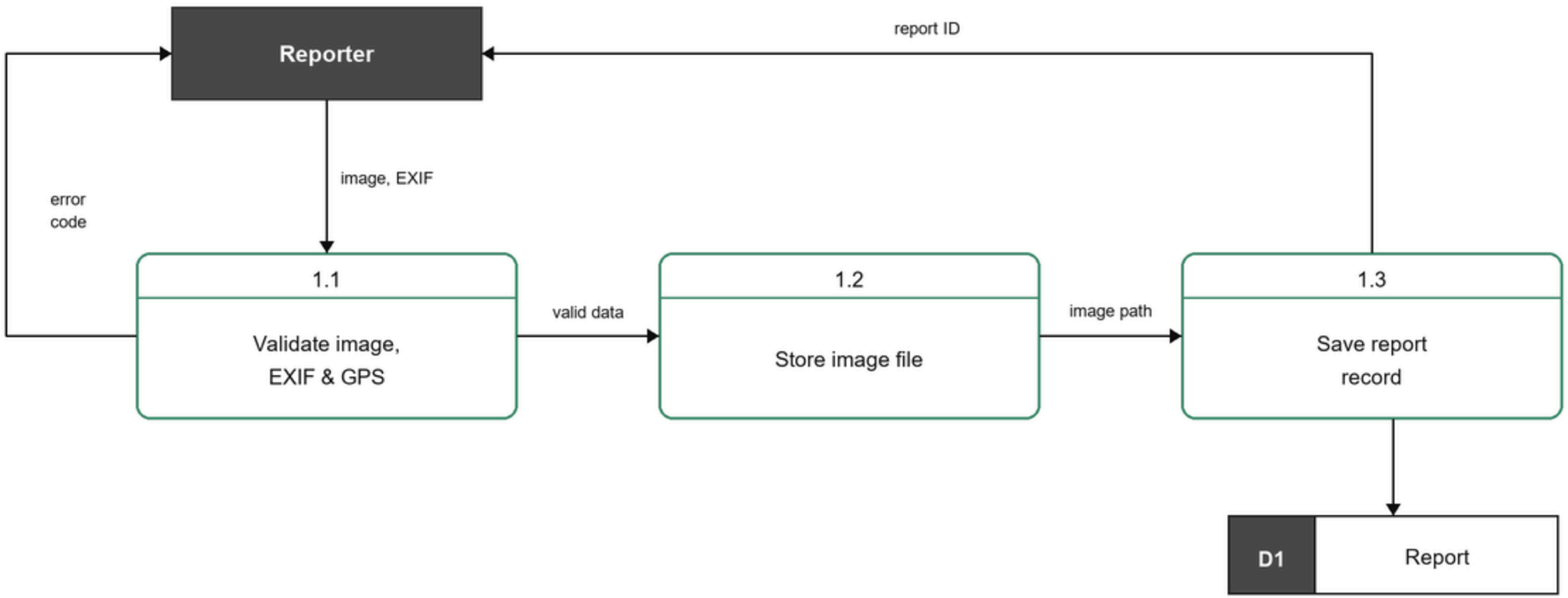
Data Flow: 6 Main System Processes seamlessly linking Reporter intake (1.0), AI processing (2.0/3.0), and Dispatch command (4.0/6.0).



State Persistence: 6 Data Stores (D1-D6) managing unified state across portals.

* Note: D6 (System & AHP Config) is a single, unified configuration store serving both MIS Dashboard (5.0) and Dispatch (6.0).

DFD Level 2: Process 1.0 Receive report



Validation Gate

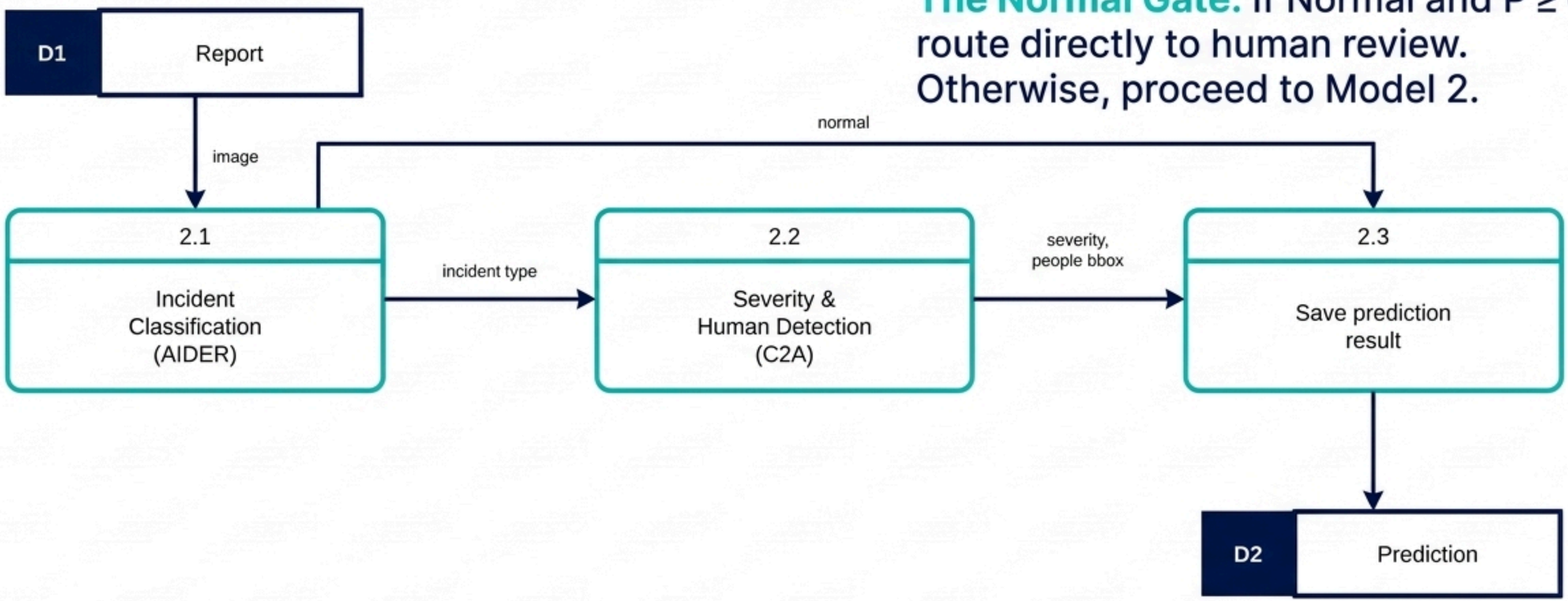
1. File validation (JPEG/PNG binaries, ≤ 35MB payload).
2. Geofence bounds (Valid EXIF GPS or AML Browser Geolocation).
3. Freshness lock (Capture timestamp ≤ 30 minutes old).
4. Integrity (SHA-256 cryptographic deduplication).

Invalid uploads return standard **ERR *** codes. Only valid binaries reach **D1** (Report Store).

Model 1 - AIDER: Scene Classifier

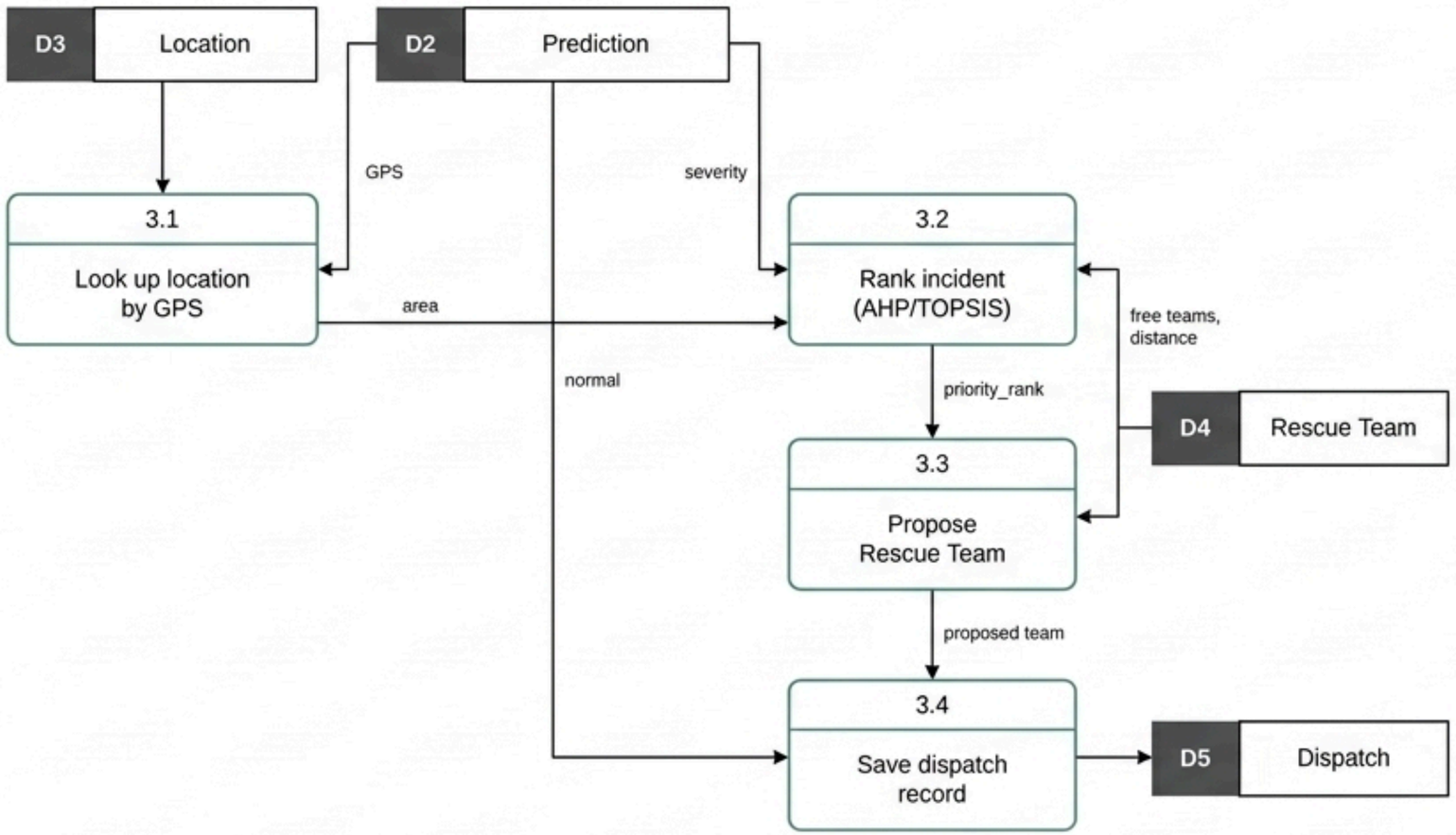
Collapsed | Fire | Flood | Traffic | **Normal**

The Normal Gate: If Normal and P ≥ 0.90, route directly to human review. Otherwise, proceed to Model 2.



Model 2 - C2A: YOLOv8 Person & Pose Detection. Detects bounding boxes and binary posture classifications (Upright vs Non-Upright). Calculates domain-specific Severity [0-100].

Gemini Notebook



Mathematical Breakdown

Step 3.1: Geolocation extraction via Haversine distance calculations.

Step 3.2: AHP/TOPSIS Multi-Criteria Queue Ranking.

Step 3.3: Base-level team eligibility matching (free status + regional coverage).

Note: ‘Proposed Team’ (3.3) is an algorithmic recommendation, not a confirmed dispatch.

Gemini Notebook

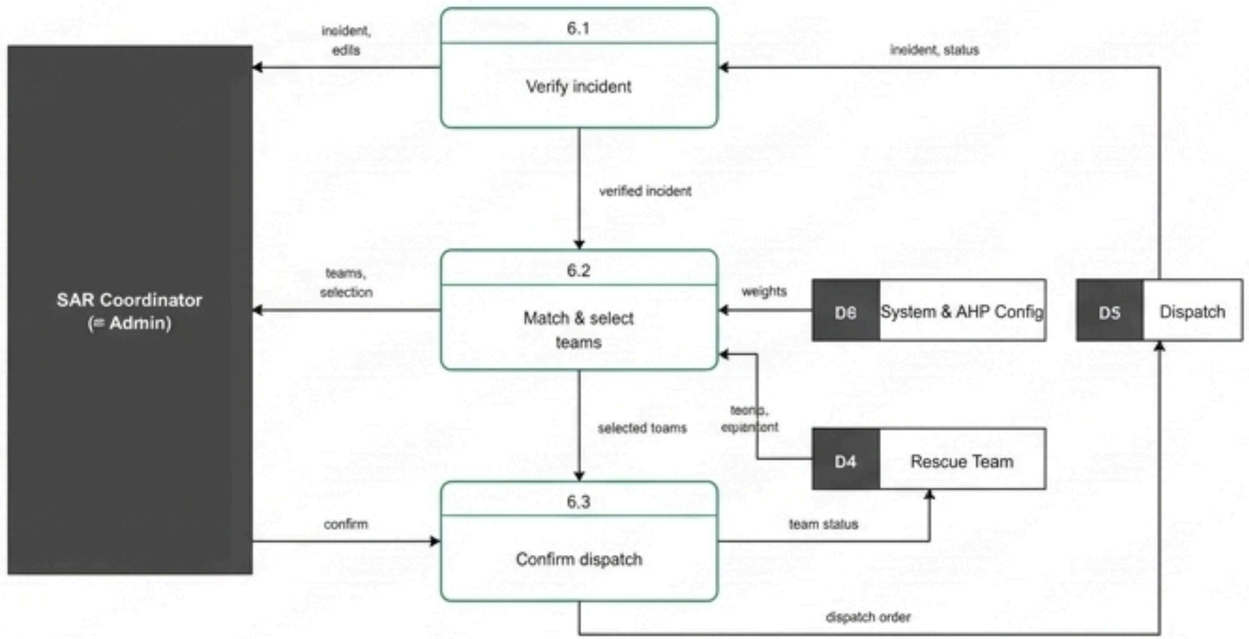
“Human-in-the-loop: AI recommends. Coordinator approves.”

6.1 Verify: Coordinator audits AI boxes and DSS Trace; holds power to override incident type, severity, and victim count.

6.2 Match: Refines AHP weights from D6 to calculate the final Team Match Score (specialty, capacity, distance).

6.3 Confirm: Generates deterministic DSP-XXXX order. Transitions team status to “On Mission”.

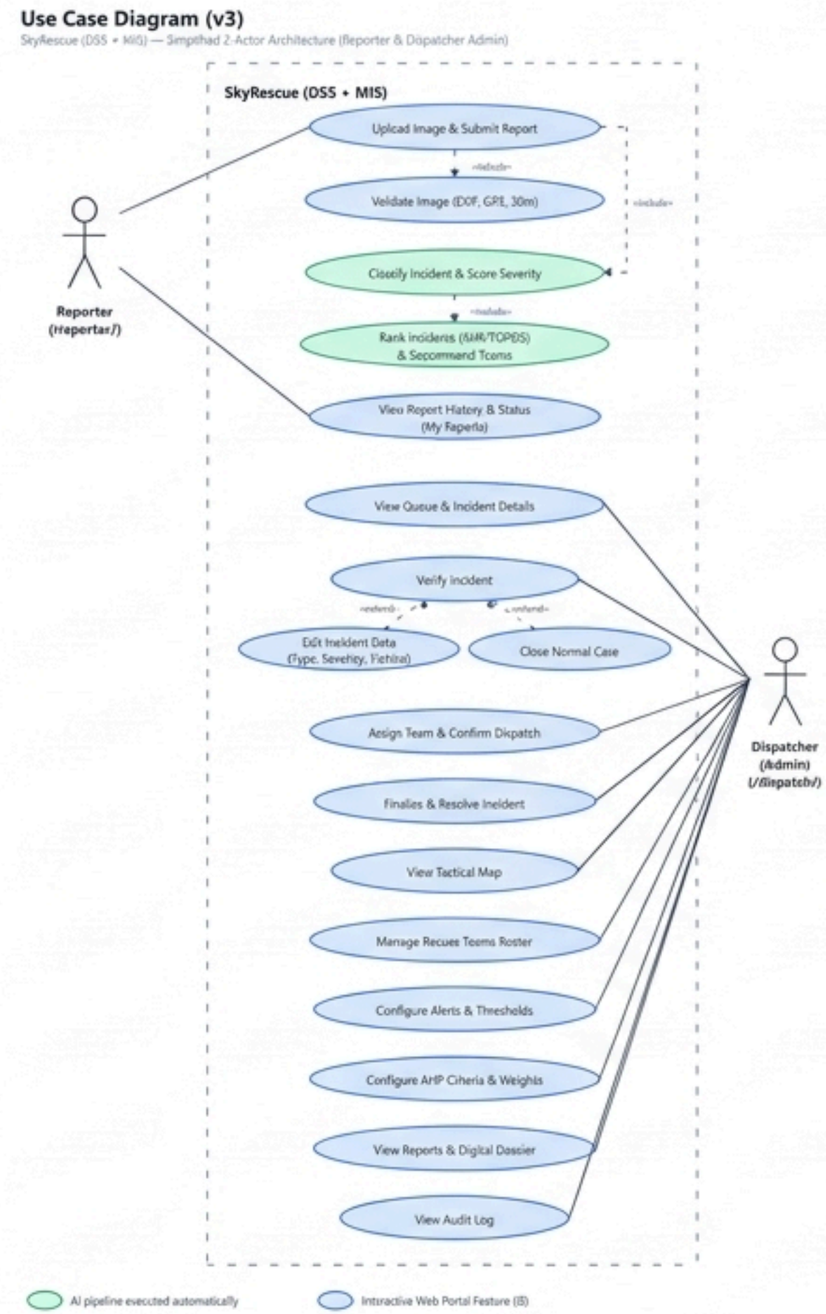
DFD Level 2: Process 6.0 Verify, match & confirm dispatch



Gemini Notebook

Reporter Portal (/reporter/)

Capabilities: Upload constraints, declaration tagging, and real-time status tracking via My Reports.



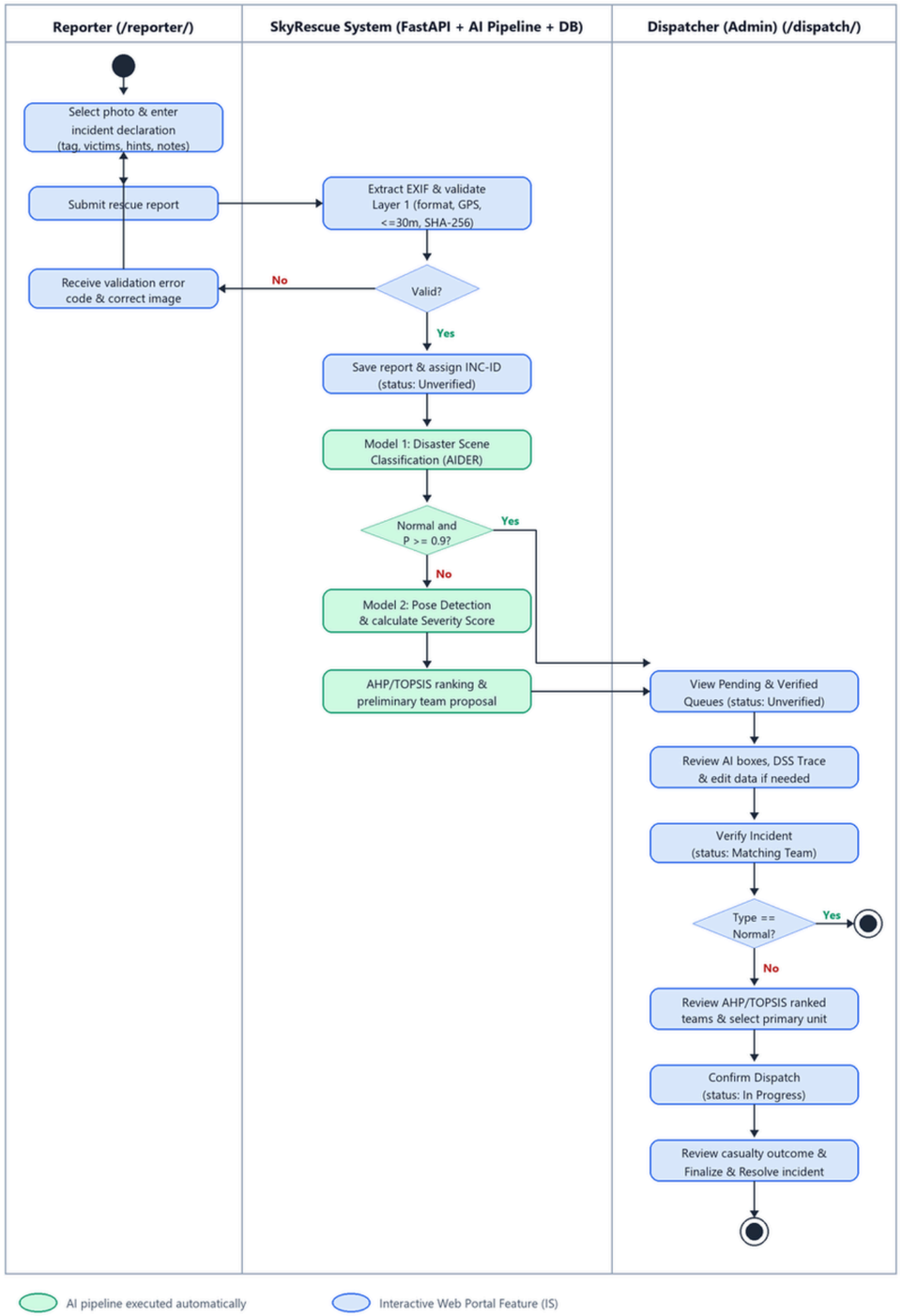
Dispatch Station (/dispatch/)

Capabilities: Queue verification, Evidence Viewer, DSS multi-criteria team assignment, tactical GIS mapping, and AHP weight configuration.

Note: Teal processes represent automated system executions requiring zero user initiation beyond upload.

Gemini Notebook

SKYRESCUE DSS: ACTIVITY DIAGRAM



Accounts

User authentication, role-based access, securely hashed session state.

Geography & Teams

Fixed location coordinates, team capacities, and M:N operational coverage matrices.

Incident & AI

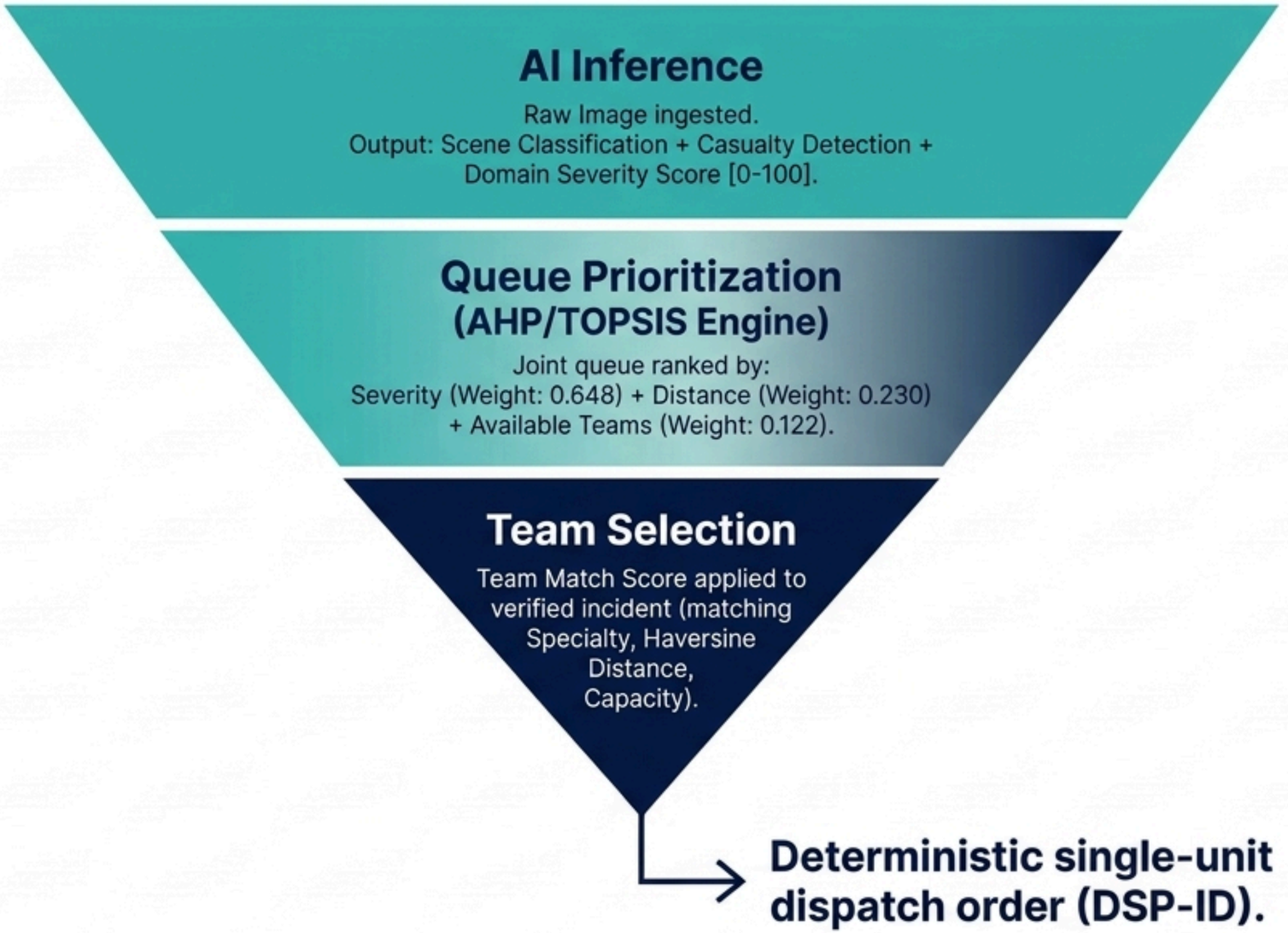
Report media payloads, AI Predictions, and 1:N Person Detection bounding boxes. Dependent variables dynamically cached.

Rules & Audit

Immutable AuditLog of all state mutations, soft-rule definitions, and upload validation attempts.

ACID-compliant SQLite database ensuring strict referential integrity (PK/FK).

Decision Intelligence Funnel





Reporter submits live aerial imagery.

Automated validation and deterministic inference execution.

Dispatcher inspects AI bounding boxes via DSS Trace and confirms the incident.

Dispatcher reviews AHP-ranked teams and confirms primary unit.

Finalizing the digital dossier and closing the loop.

Demo Scenario: Concurrent processing of one High-Severity incident and one P > 0.90 Normal incident.

Gemini Notebook

Data to Evidence

Computer Vision (AIDER/C2A) converts raw, unstructured aerial pixels into structured, measurable severity metrics.

Mathematical Triage

AHP/TOPSIS decision-support mathematically guarantees critical hazards receive attention first.

Shared Truth

Real-time MIS sync and immutable audit trails ensure perfect operational symmetry between field and command.

Human Command

The system never replaces the SAR Coordinator; it empowers them with speed, precision, and explainable evidence.

SkyRescue: From Aerial Imagery to Coordinated Response.